

Spring 2026

Multi-agent Sampling Drone Project

Sprint 3 Review (4/10)

Guideline to creating your slides

- Goals & Progress:
 - Use bullet points to highlight the Sprint goals and what was achieved
 - **Use photos and videos over blocks of text**
- Next Steps: Highlight your plans for the next sprint
- Each team should plan to have 2-5 slides showcasing their work (don't go overboard)

Stream 1 – Drone Software

End-of-Semester Goals

- 3 drones flying autonomously
- End-to-end sampling pipeline
- Precise positioning of sampler w/ camera
- Realistic sim environment for testing
- Improve custom mounting of parts

Sprint 3 Goals

- Mount all custom parts
- Implement custom landing sequence
- End to end test in sim with slung load
- Confirm flight capability with slung load
- End to end test with real drone

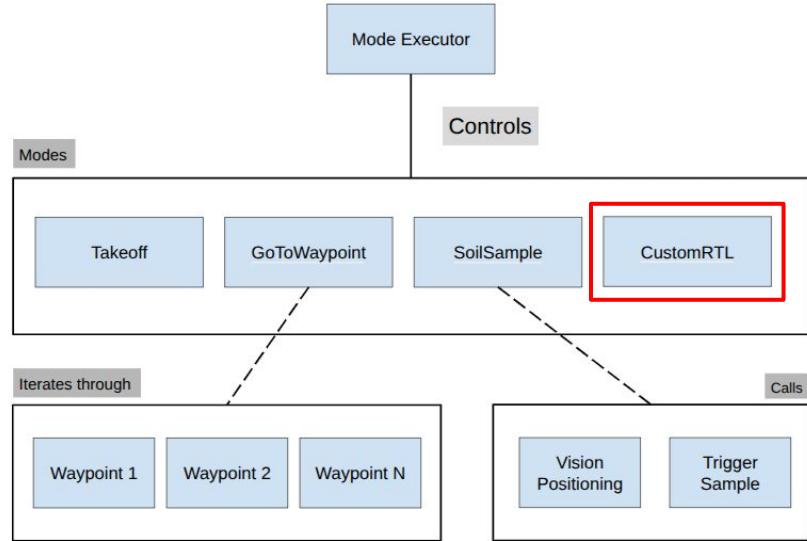
Sprint 3 Goals

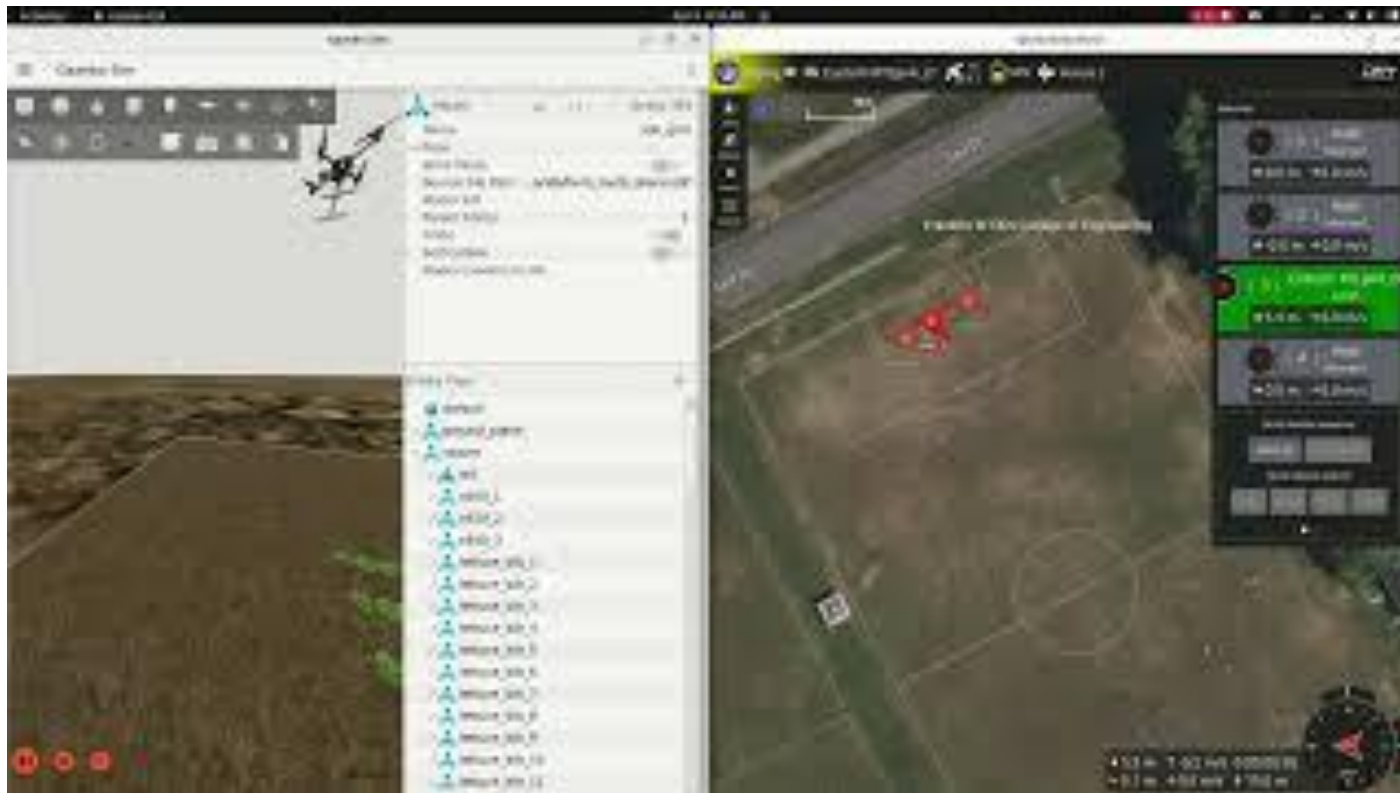
- ~~Mount all custom parts~~
- ~~Implement custom landing sequence~~
- ~~End to end test in sim with slung load~~
- ~~Confirm flight capability with slung load~~
- End to end test with real drone (in progress)

Custom landing sequence

- Implemented custom landing mode
 - Descend to land sampler
 - More forward
 - Land fully

Control Overview





Custom RTL, simulated

Flight with slung load

Weight is fine!!!



Full mission test



Full mission, real drone

Full mission test

- **No slung load** at time of testing
 - Will be testing at 5pm today
- **No CAN** connection with soil sampler
 - Currently replaced with 5 sec hold
- **No camera-based** position readjustment
 - Most likely out of scope for this semester

Plan for Next Sprint

- Set up camera for video capturing
- Integrate soil sampler actuation to ROS2 code

Stream 1 – Sampler System Integration

Including interconnection unit, sensor and actuator integration

End of Semester Goals

- Finalize communication for driving the servos and sending the sensor readings to the NUC
- Integrate servos and sensors to detect probe extension with the Nano on board the sampler
- Complete sampling routine that the NUC can control
- Fully test soil sampler and drone integration

Sprint 3 Goals

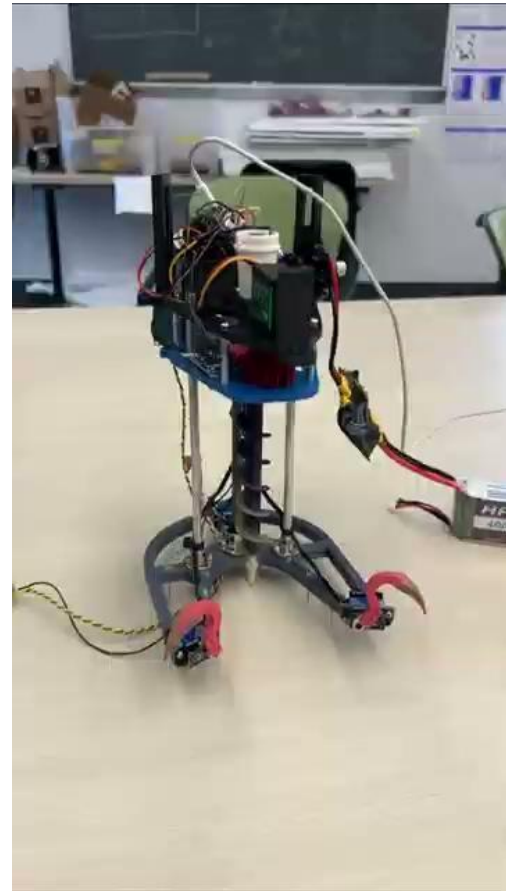
- Calibration of soil sensor
- NUC can communicate over CAN
- CAN communication works over 2 meters in length
- Full servo sampling routine can be initiated by a Python script on the NUC

Sprint 3 Goals

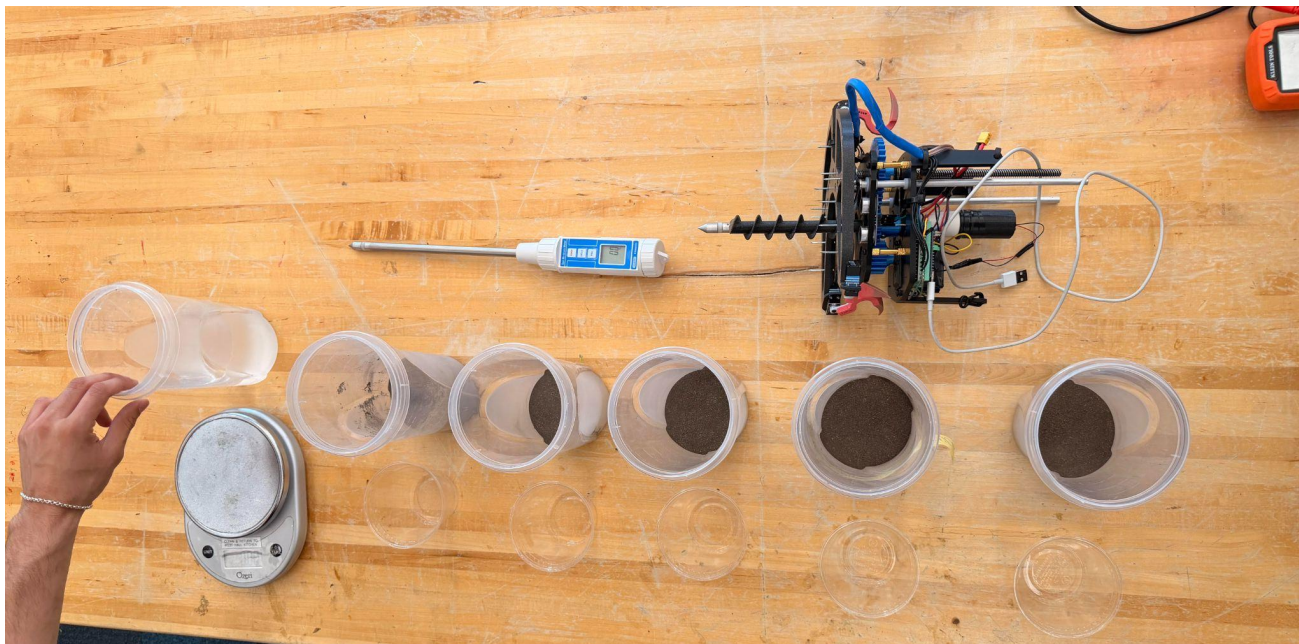
- Calibration of soil sensor
- ~~● NUC can communicate over CAN~~
- ~~● CAN communication works over 2 meters in length~~
- Full servo sampling routine can be initiated by a Python script on the NUC
 - Mostly there – haven't tested full routine with distance sensor + issue with hanging when repeating sampling

Sampler control over CAN

- All servos can be driven either off of Serial (for debugging) or CAN signals from C++ code (when connected to the battery)
- Sampling routine can execute a single loop (with fixed extension time) and send data back over CAN
 - Hangs when sending data a second time — possibly some overflow? Looking into it
 - Reading measurements from the distance sensor, but haven't tested with these yet



Soil Calibration



Plan for Next Sprint

- By end of weekend (Sunday):
 - Fix hanging bug when running the sample routine for the second time
 - Test using distance reading for extension/retraction
- Verify routine behaves correctly in ROS code
- Verify consistency of sensor readings when in soil, during sampling

Stream 1 – Mission Planning

Assumption 1: Complete Duty Cycle before Battery Life ends

1) Define # cells based on Assumption 1

Args:

- Polygon area
- Sampling Time
- Flight Speed
- Total Mission Time
- Number of Agents

2) Voronoi Iterations/
Lloyd's Algorithm for
generating equal cells

Args:

-

3) Vehicle Routing
Problem

- Traveling
Salesman
re-imagined

4) Time delays to
avoid route
collisions

```
1  "px4_1/": {  
2    "waypoints": [  
3      {  
4        "lat": 42.292699,  
5        "lon": -71.262827,  
6        "alt": 3.0,  
7        "type": "depot"  
8      },  
9      {  
10       "lat": 42.292588,  
11       "lon": -71.262906,  
12       "alt": 3.0,  
13       "type": "depot"  
14     }  
15   ]  
16 }
```

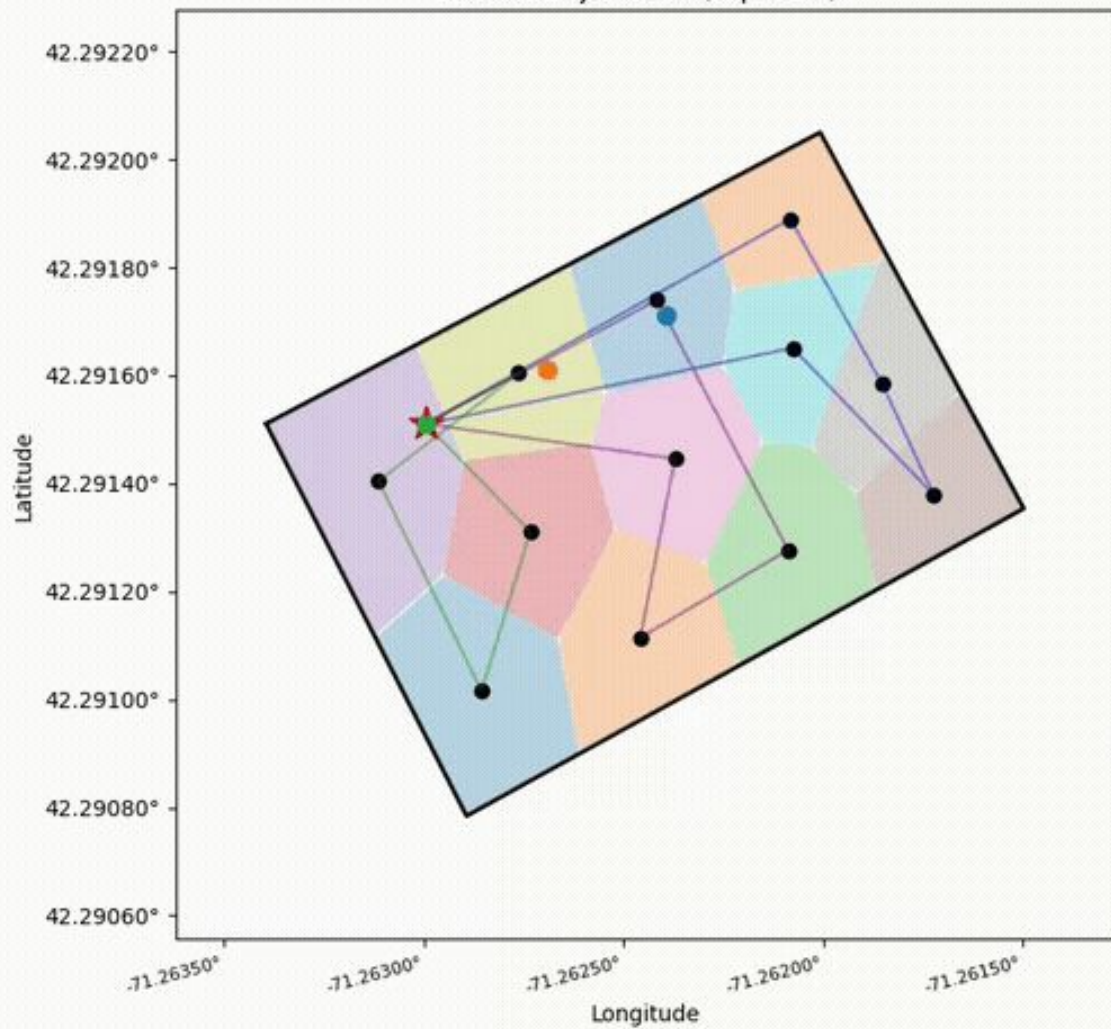
Json file with lat, long, alt, type of way point

Sprint 3 Updates

- Debugged VRP to work as intended
- Added timed delays → d-safe variable determines the range of collision to avoid and account for
- Updated json to include information like “sampling point” or “depot” to toggle different hover time periods
- Tested in simulation
 - a. Changed depot points to be on the field



Drone Trajectories (Top View)



Sprint 3 Next Steps

- Testing range of time delays to ensure no collisions even if set up time is off
- Updates to json to include delays for collision avoidance

Future goal:

- Re-charge/battery swap path planning
- Dashboard

Stream 2 – Soil Sampler Development

End-of-Semester Goals

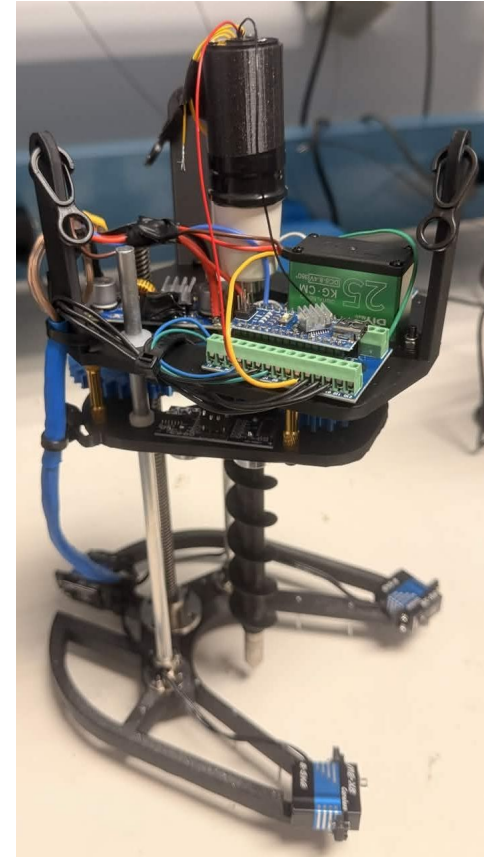
- Fully Functional Soil Sampler x 3
 - Position detection on Soil Sampler
 - Land without tipping
 - Drill all the way through without anchors slipping
 - Measure the distance the soil sampler travelled

Sprint 3 Goals

- Finalize the Design & 1 fleshed out model
- Buy enough parts for all 3 Soil Sampler design
- 3D printed all parts
- Look into nylon lead screw

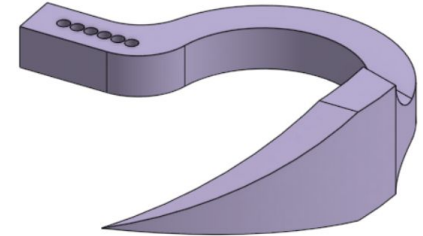
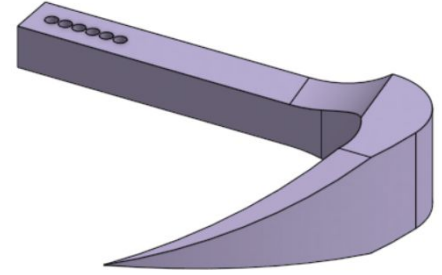
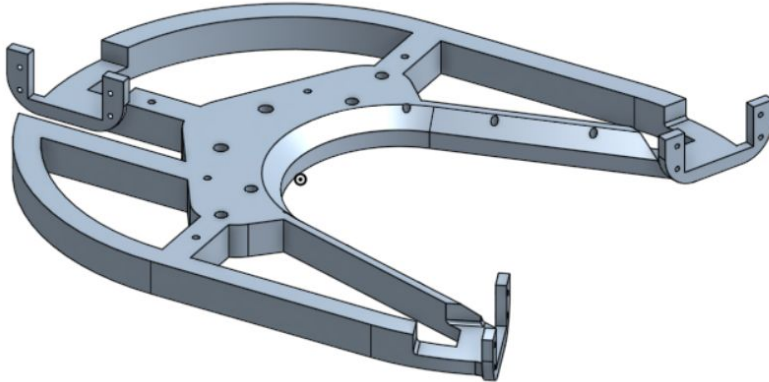
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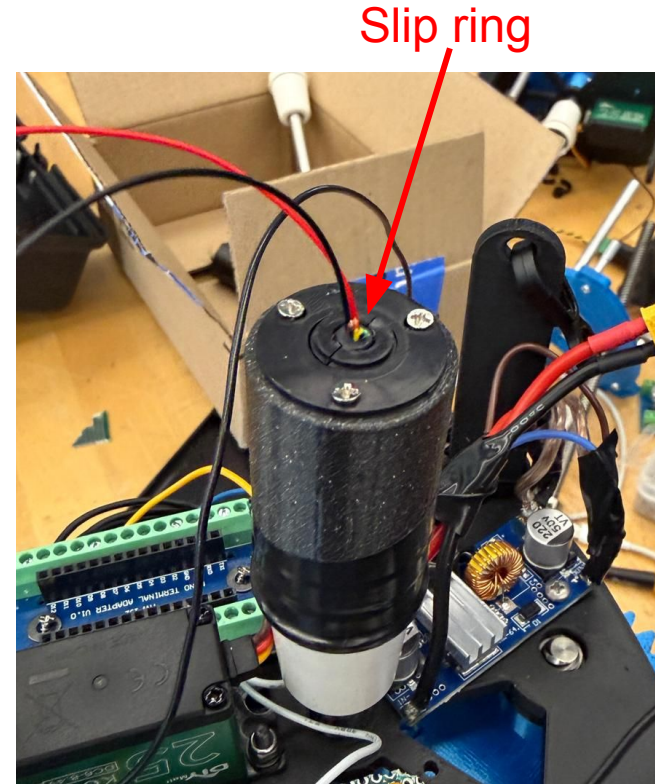
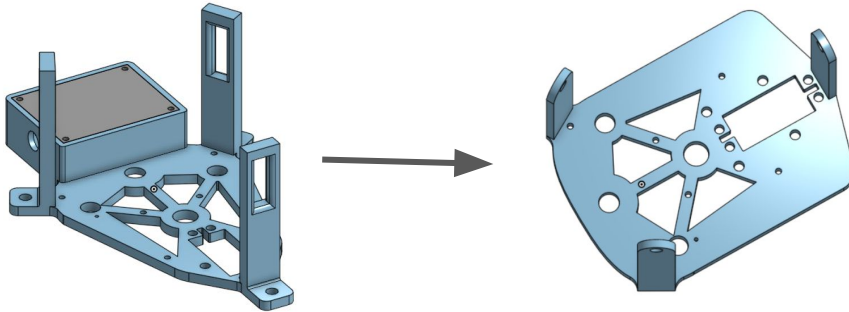
Sprint Progress (1/4)

- **Anchors Redesign & Base changes**
 - Shorter & Round anchors
 - Open base design



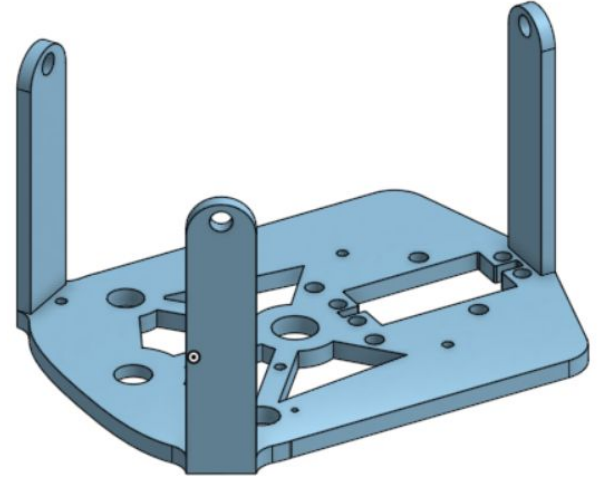
Sprint Progress (2/4)

- **Probe Head Holder + Electrical**
 - Probe Holder assembled
 - Electrical Rewired and packaged
 - Probe is fully wired



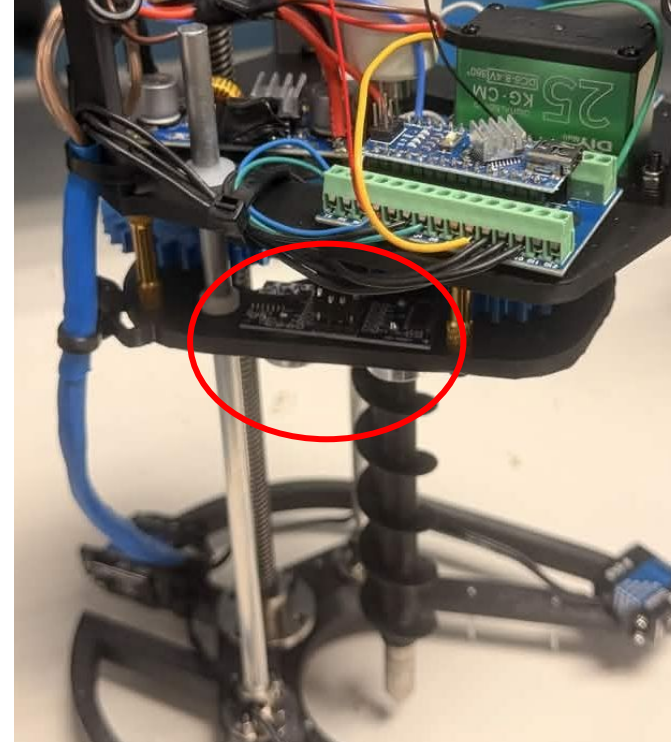
Sprint Progress (3/4)

- **Drone & Soil Sampler Attachment**
 - The attachment holes are on top rather than the ones on the same level as the plate

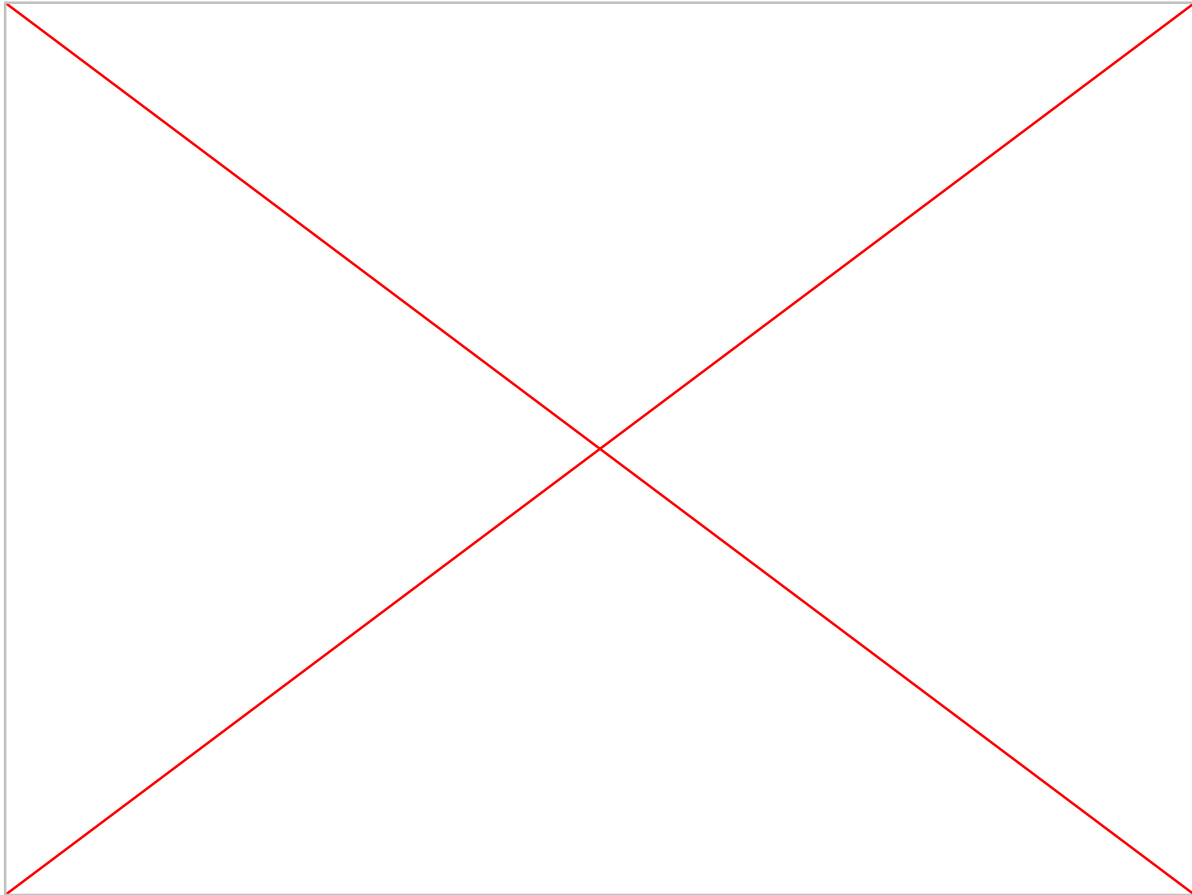


Sprint Progress (4/4)

- **Distance measurement**
 - Ultrasonic sensor mounted, wired, and functional

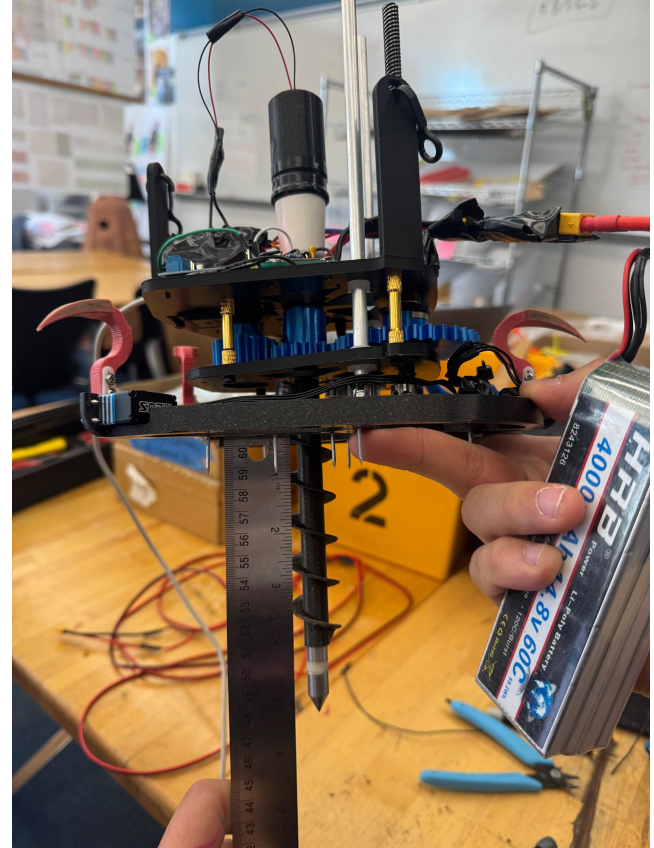


Soil Sampler working



Soil Sampler details

- **Distance**
 - Can go a little bit more than 5 inches
 - Weight: 694 grams



Challenges Faced

- **Anchors don't work for all types of soil:** need to do more testing to figure out which soil compositions/densities it works for
- **Wire tangling:** try attaching the ropes directly to the attachment points, without the clips
- **Servo mounts on base breaks easily:** add more material for strength

Plan for Next Sprint

- Finish assembling last sampler
- Fully wire the first sampler
- Finish testing anchor design on different soils
- Test tangling of wires & potentially reiterate attachment design

Project Timeline

- **May 3** **FRC submission deadline** (report and video)
- *May 2* *Complete FRC submission*
- **May 1** **Sprint 4 review** (end of academic year celebration)
- *Apr 30 - May 1* *Report review and revision*
- *April 20 - 29* *Final documentation and report writing & video editing*
- *April 11 - 19* *Final testing and video recording*
- **April 10** **Sprint 3 review**
- *March 27* *Start writing final report & video planning*
- *Mar 23 - Apr 10* *Final design and implementation push*
- **March 13** **Sprint 2 review**

Project Timeline